

Solution Requirements

Date	06 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

A comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the application using a login system or authenticated session before starting assessments.	High
2	Authorization levels	The system supports role-based access where students can take assessments and faculty can review reports and performance results.	High

3	External interfaces	The application integrates with Speech-to-Text APIs (OpenAI Whisper or SpeechRecognition), Sentence-BERT for semantic similarity, Streamlit UI, and PDF report generation libraries.	High
4	Transactions processing	The system records the user's spoken response, converts it into text, evaluates conceptual understanding, calculates the score, and stores the results.	High
5	Reporting	The application generates concept understanding scores, semantic similarity results, feedback, visual summaries, and downloadable PDF reports.	High
6	Business rules, etc.	Only valid audio responses are processed. Scores are calculated based on semantic similarity and predefined evaluation criteria. Invalid or empty inputs are rejected with appropriate messages.	High
7	Compliance to laws or regulations	The system maintains user privacy by securely handling uploaded audio and generated reports without exposing personal data.	Medium

8	<i>Other</i>	The system provides an intuitive interface for recording/uploading audio, viewing results, and downloading reports for future reference.	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should process speech, evaluate responses, and display results quickly.	Complete evaluation within 5–10 seconds after submission.
2	Scalability	The system should support multiple assessments and increasing numbers of users without significant performance degradation	Supports 100+ assessment sessions with stable performance.
3	Security & Data Privacy	User audio files and evaluation reports should be securely handled and protected from unauthorized access.	Secure storage, authenticated access, and no unauthorized data exposure.
4	Reliability & Availability	The application should function consistently with minimal failures during assessments.	99% availability during normal operation.
5	Usability & Accessibility	The interface should be simple, responsive, and easy for students and faculty to use with minimal training	Users should be able to complete an assessment in 3 simple steps.

6	Other (Maintainability & Portability)	The system should be easy to update, maintain, and deploy on different environments using Python and Streamlit.	Modular codebase with support for deployment on Windows, Linux, and cloud platforms.
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